Dhyey Patel

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Education

M.S. Computer Software Engineering GPA: 3.78/4.0
Arizona State University, Tempe, AZ *Jan 2025 – Dec 2026*
Courses: *Cloud Computing, Advanced Data Structures and Algorithms, Data Processing at Scale, AI*

B.E. Information Technology Engineering GPA: 9.21/10.0
Gujarat Technological University, Ahmedabad, Gujarat, India *Oct 2020 – Jun 2024*
Courses: *Machine Learning, Artificial Intelligence, OOP, Computer Vision*

Technical Skills

Languages: Python, C++, Java, SQL, TypeScript, JavaScript
Systems & Backend: Linux, FastAPI, Spring Boot, Node.js, RESTful APIs, Celery
Cloud & Distributed Systems: AWS, Docker, Kubernetes, Lambda, SQS, Step Functions, DynamoDB, S3
Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis
Frontend: React, HTML5/CSS3
DevOps & Testing: Git, GitHub Actions, Jenkins, CI/CD, Jest, Mockito, Postman
AI/ML: PyTorch, TensorFlow, Transformers, AWS Bedrock, Scikit-learn

Experience

Software Development Engineer Intern, Amazon.com, Inc. May 2026 – Aug 2026
Amazon Music *San Francisco, CA, USA*

- Built and deployed a production Python-based decision platform processing **17.7K+ catalog decisions/day** at **99% precision**, reducing resolution time from days to 90 seconds.
- Engineered fault-tolerant processing with automated retries, failure isolation, audit logging, and production monitoring, achieving zero failures across **142K+ production decisions**.
- Architected distributed event-driven workflows using AWS Lambda, Step Functions, SQS, and DynamoDB, optimizing batching and concurrency to sustain 740 decisions/hour while preventing downstream resource throttling.
- Collaborated with Applied Scientists and Software Engineers through design, testing, and code reviews to identify system bottlenecks and design a scaling **path from 17.7K to 88K+ decisions/day**.

Full-Stack Developer Intern, Xmatiq Technologies Jan 2024 – Apr 2024
Techforever Solutions LLP. *Ahmedabad, GJ, India*

- Developed backend services and RESTful APIs for ERP systems using Node.js, Express.js, and MongoDB, designing data models and service integrations that improved data processing **efficiency by 80%.**
- Built an end-to-end web application using PHP and MySQL, developing RESTful APIs with input validation, error handling, and third-party payment integration for reliable production use.

ReactJS Intern Jul 2023 – Aug 2023
Ingenious Technologies *Vadodara, GJ, India*

- Implemented secure RESTful APIs with JWT authentication for job listings and candidate data, handling **200+ requests/minute** with **99% uptime** while improving backend reliability.
- Refactored reusable ReactJS components by introducing **lazy loading** and optimized state management, reducing page load times by **30%** and improving application performance and maintainability.

Projects

Cloud-Native Microservices Platform Aug 2025 – Dec 2025

- Deployed containerized Python services on AWS EKS using Docker, Kubernetes, Terraform, and Helm, building reproducible infrastructure across multiple environments.
- Implemented CI/CD and Prometheus/Grafana monitoring for automated deployments, service health tracking, and operational visibility across distributed services.

NextMeet-AI Meeting Companion Feb 2025 – Jun 2025

- Built a full-stack AI meeting companion using React, FastAPI, PostgreSQL, Redis, and Celery, developing responsive interfaces and REST APIs for transcription, summarization, and action-item workflows.
- Designed asynchronous processing with Celery and Redis to decouple long-running AI workloads from API requests, providing real-time task progress and responsive frontend updates.

Event-Driven Microservices Order Management System Jan 2026 – Mar 2026

- Architected a Python-based **event-driven system** with independently scalable order, inventory, and analytics services using AWS Lambda, SQS, API Gateway, and DynamoDB for asynchronous communication.
- Implemented idempotent consumers, queue-based decoupling, authentication, and failure handling to maintain data consistency and service reliability across distributed workflows.

EXTRACURRICULAR ACTIVITIES

- **3rd Place - Intel Hack the Vision 2025**: Contributed to an open-source project for Intel's AI-powered chipset.